

LIF Action Mapping to Audio and MIDI Output

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1 Overview

This document specifies how neural activity from a leaky integrate-and-fire (LIF) network is converted to:

1. tone synthesis output, and
2. MIDI note output.

The runtime pipeline has three logical stages:

- **Input layer:** audio features (8-band spectrum and RMS) are mapped to neuron drive.
- **Hidden layer:** recurrent LIF neurons evolve according to topology-specific connectivity.
- **Output layer:** a sampled activity column is mapped to 16 bins, then to audio oscillators and MIDI notes.

2 Neural Structure and Equations

2.1 Input Layer: Audio-to-Neuron Drive

Let $b_g \in [0, 1]$ be audio band energy for group $g \in \{0, \dots, 7\}$ and $r \in [0, 1]$ be RMS. Each neuron i is assigned a band index:

$$g(i) = \left\lfloor \frac{8i}{N} \right\rfloor \quad (1)$$

where N is neuron count.

With influence control $\alpha \in [0, 1]$, define:

$$\gamma(\alpha) = 0.35 + 1.65\alpha, \quad (2)$$

$$u_i = \gamma(\alpha) \left[b_{g(i)} (0.30 + 1.70r) + 0.15 b_{(g(i)+3) \bmod 8} \right]. \quad (3)$$

Here u_i is the external input current to neuron i .

2.2 Hidden Layer: LIF Dynamics

Each neuron keeps membrane potential v_i , spike flag s_i , and refractory timer. Using per-frame step Δt , parameters vary with influence:

$$\lambda = 0.70 - 0.45\alpha \quad (\text{leak}), \quad (4)$$

$$\theta = 0.75 - 0.35\alpha \quad (\text{threshold}), \quad (5)$$

$$v_{\text{reset}} = 0.04 + 0.10(1 - \alpha), \quad (6)$$

$$\tau_{\text{ref}} = 0.03 + 0.08(1 - \alpha). \quad (7)$$

For weighted adjacency matrix $W = [w_{ji}]$:

$$I_i(t) = u_i(t) + \sum_{j=1}^N w_{ji} s_j(t). \quad (8)$$

A discrete LIF update is modeled as:

$$v_i(t + \Delta t) = \begin{cases} v_{\text{reset}}, & \text{if refractory is active,} \\ \text{clip}((1 - \lambda \Delta t)v_i(t) + \Delta t I_i(t), 0, 1), & \text{otherwise.} \end{cases} \quad (9)$$

Spike event:

$$s_i(t + \Delta t) = \begin{cases} 1, & v_i(t + \Delta t) \geq \theta, \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

On spike, $v_i \leftarrow v_{\text{reset}}$ and refractory timer is set to τ_{ref} .

2.3 Output Layer: Column Sampling to 16 Bins

At scan phase $\phi \in [0, 1]$, choose spatial column:

$$x = \text{round}(\phi(G - 1)) \quad (11)$$

where G is LIF grid size.

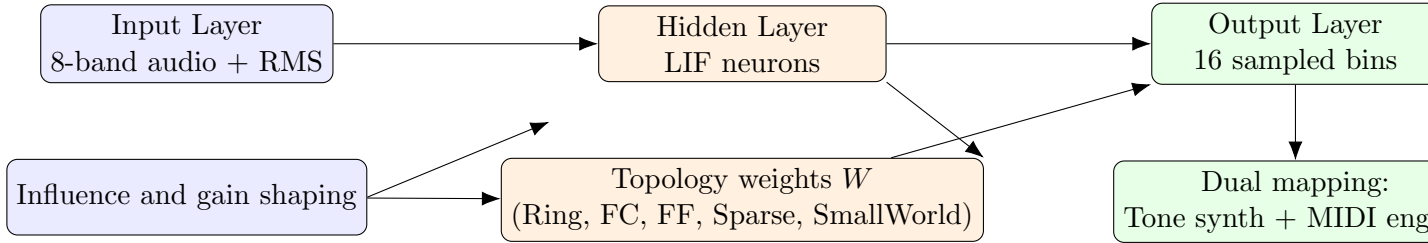
For output bin $k \in \{0, \dots, 15\}$:

$$y_k = \text{round}\left(\frac{k + 0.5}{16}(G - 1)\right), \quad (12)$$

$$e_k = \text{clip}(0.8 v_{x, y_k} + 0.6 s_{x, y_k}, 0, 1). \quad (13)$$

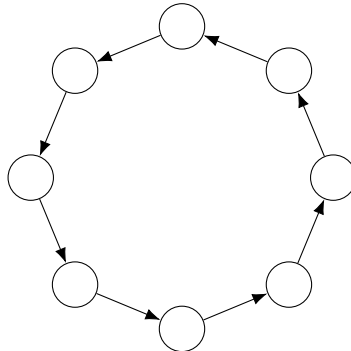
The vector $\mathbf{e} = (e_0, \dots, e_{15})$ is the shared output for audio and MIDI mapping.

3 Layered Signal-Flow Diagram



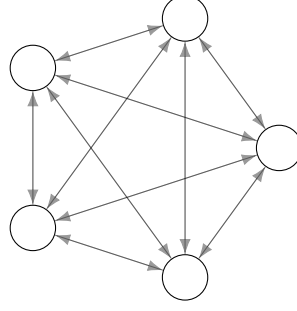
4 Topology Diagrams (Hidden Layer Connectivity)

4.1 Ring Topology



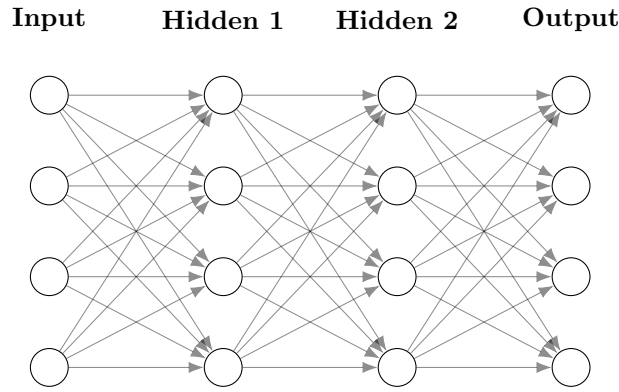
Local cyclic neighbor coupling (plus multi-hop links in implementation).

4.2 Fully Connected Topology



All-to-all recurrent coupling (excluding self loops).

4.3 Feedforward Topology (Layer-Separated)



4.4 Sparse Random and Small-World Topologies

Topology	Connectivity idea
Sparse Random	Each directed edge exists with low probability (about 10%).
Small-World	Mostly local ring-like edges plus occasional rewired shortcuts.

5 Algorithm Pseudocode

Algorithm 1 Input Layer: Audio Features to Neuron Drive

Require: Bands $b[0..7]$, RMS r , influence α , neuron count N

```

1: for  $i = 0$  to  $N - 1$  do
2:    $g \leftarrow \lfloor 8i/N \rfloor$ 
3:    $\text{bandDrive} \leftarrow b[g] \cdot (0.30 + 1.70r)$ 
4:    $\text{cross} \leftarrow 0.15 \cdot b[(g + 3) \bmod 8]$ 
5:    $\gamma \leftarrow 0.35 + 1.65\alpha$ 
6:    $u[i] \leftarrow (\text{bandDrive} + \text{cross}) \cdot \gamma$ 
7: end for
```

Algorithm 2 Hidden Layer: LIF Network Step

Require: Previous state (v, s, ref) , input u , weights W , parameters $(\lambda, \theta, v_{\text{reset}}, \tau_{\text{ref}})$

```
1: for  $i = 0$  to  $N - 1$  do
2:    $I \leftarrow u[i] + \sum_j W[j, i] \cdot s[j]$ 
3:   if  $\text{ref}[i] > 0$  then
4:      $\text{ref}[i] \leftarrow \text{ref}[i] - \Delta t$ 
5:      $v[i] \leftarrow v_{\text{reset}}$ 
6:      $s[i] \leftarrow 0$ 
7:   else
8:      $v[i] \leftarrow \text{clip}((1 - \lambda \Delta t)v[i] + \Delta t I, 0, 1)$ 
9:     if  $v[i] \geq \theta$  then
10:       $s[i] \leftarrow 1$ 
11:       $v[i] \leftarrow v_{\text{reset}}$ 
12:       $\text{ref}[i] \leftarrow \tau_{\text{ref}}$ 
13:    else
14:       $s[i] \leftarrow 0$ 
15:    end if
16:  end if
17: end for
```

Algorithm 3 Output Layer: Column Sampling to 16 Energy Bins

Require: LIF grid state (v, s) , scan phase $\phi \in [0, 1]$

```
1:  $x \leftarrow \text{round}(\phi(G - 1))$ 
2: for  $k = 0$  to  $15$  do
3:    $y \leftarrow \text{round}(((k + 0.5)/16) \cdot (G - 1))$ 
4:    $e[k] \leftarrow \text{clip}(0.8 \cdot v[x, y] + 0.6 \cdot s[x, y], 0, 1)$ 
5: end for
6: return  $e[0..15]$ 
```

Algorithm 4 Audio Mapping: Bins to Additive Oscillator Bank

Require: Energies $e[0..15]$, min/max frequency $(f_{\text{min}}, f_{\text{max}})$, sample rate F_s

```
1: for  $k = 0$  to  $15$  do
2:    $t \leftarrow k/15$ 
3:    $f[k] \leftarrow f_{\text{min}} \cdot (f_{\text{max}}/f_{\text{min}})^t$  ▷ log-frequency spacing
4: end for
5: for each audio frame do
6:    $x \leftarrow 0$ 
7:   for  $k = 0$  to  $15$  do
8:      $a[k] \leftarrow a[k] + 0.01 \cdot (e[k] - a[k])$  ▷ amplitude smoothing
9:      $\varphi[k] \leftarrow \varphi[k] + 2\pi f[k]/F_s$ 
10:     $x \leftarrow x + \sin(\varphi[k]) \cdot a[k]$ 
11:  end for
12:  output sample  $\leftarrow \text{clip}(0.5x/16, -1, 1)$ 
13: end for
```

Algorithm 5 MIDI Mapping: Bins to Notes and Velocity

Require: Energies $e[0..15]$, style, modal scale, key, note range

```
1:  $e_{max} \leftarrow \max_k e[k]$ 
2:  $\tau_{on} \leftarrow \max(0.18, 0.55 e_{max})$ 
3:  $\tau_{off} \leftarrow \max(0.12, 0.40 e_{max})$ 
4: for  $k = 0$  to 15 do
5:   compute harmonic function  $F_k \in \{T, S, D\}$  from bin index and local contour
6:   choose chord notes from  $(F_k, \text{style}, \text{modal scale}, \text{key})$ 
7:   map notes into configured MIDI range
8:   if feedforward topology then
9:     use pulse gate + cooldown scheduling
10:    velocity  $\leftarrow \text{clip}(30 + 97 e[k]^{0.7} + 10 e[k-1], 30, 127)$ 
11:   else
12:     if note is OFF and  $e[k] > \tau_{on}$  then
13:       velocity  $\leftarrow \text{clip}(20 + 107 (e[k] / \max(e_{max}, \epsilon))^{0.65}, 20, 127)$ 
14:       send NoteOn for chord notes
15:     else if note is ON and  $e[k] \leq \tau_{off}$  then
16:       send NoteOff for active notes
17:     end if
18:   end if
19: end for
```

6 Audio and MIDI Output Mapping Summary

Signal	Audio output	MIDI output
Bin index k	Oscillator index/frequency channel	Harmonic role and register
Bin energy e_k	Oscillator amplitude	Gate and velocity
Scan phase ϕ	Time-varying timbre source	Time-varying note activity
Topology	Activity texture and transients	Sustained vs pulsed articulation

7 Notes on Layer Differentiation

- **Input layer** contains externally observed features only (audio bands, RMS, influence controls).
- **Hidden layer** contains recurrent neural state and topology-defined coupling.
- **Output layer** is the 16-bin activity vector, consumed by both audio synthesis and MIDI generation.